

4.5. Facilities Design

The seven management and planning tools are the

- 1. Affinity diagram,**
- 2. The interrelationship digraph,**
- 3. The tree diagram,**
- 4. The matrix diagram,**
- 5. The contingency diagram,**
- 6. The activity network diagram, and**
- 7. The prioritization matrix**

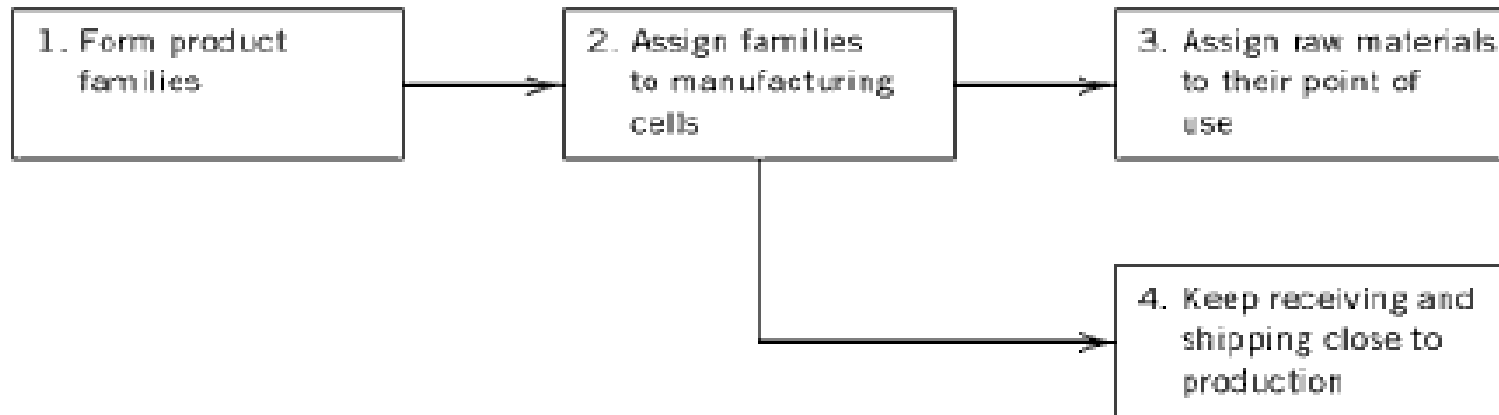
Issues in reducing manufacturing lead time

Facilities design	Equipment issues	Quality	Setup time	Scheduling
1. Form product families	1. Operator certification program	1. Provide training on how to use process documentation	1. Provide documentation on setup procedures	1. Provide visibility to daily product sequence
2. Assign families to cells	2. Sit technicians closer to production	2. Implement successive inspection with feedback	2. Locate fixtures and tooling close to machines	2. Do not authorize products for which the needed parts are not available
3. Assign raw mat's to their point of use	3. Monitor breakdowns to predict future occurrences	3. Develop mistake-proof devices	3. Provide training so operators can participate	3. Negotiate frequent and smaller lots to customers
4. Keep receiving and shipping close to production	4. Recruit enough technicians per shift	4. Develop capabilities for monitoring key machine parameters	4. Provide information on daily sequence	

Affinity diagram example for reducing manufacturing lead time

2.5.2 Interrelationship Diagram

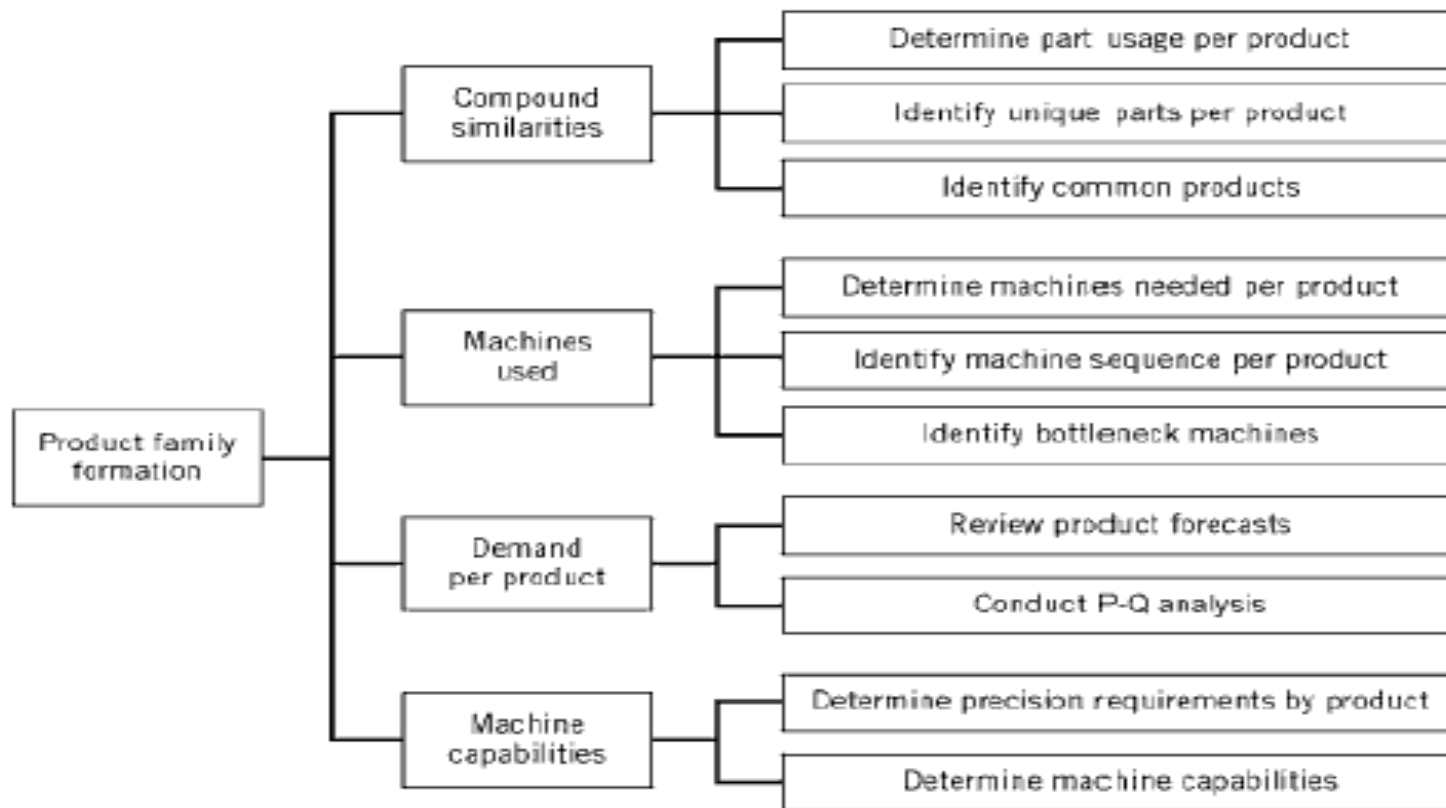
- The interrelationship diagram is used to map the logical links among related items, trying to identify which items impact others the most. This graph helps us understand the logical sequence of steps for the facilities design



Interrelationship diagram for facilities design

2.5.3 Tree Diagram

- The tree diagram is used to map in increasing detail the actions that need to be accomplished in order to achieve a general objective.



Tree diagram for the formation of product families

2.5.4 Matrix Diagram

The matrix diagram organizes information such as characteristics, functions, and tasks into sets of items to be compared. This tool provides visibility to key contact on specific issues and helps identify individuals who are assigned to too many teams.

Team\Participants	Joe	Mary	Jerry	Lou	Linda	Daisy	Jack
Part usage team	P	C	P	L			P
Machine use & cap team	L		C				P
Demand forecast team				P	C	L	

Note. L: Team Leader

C: Team Coordinator

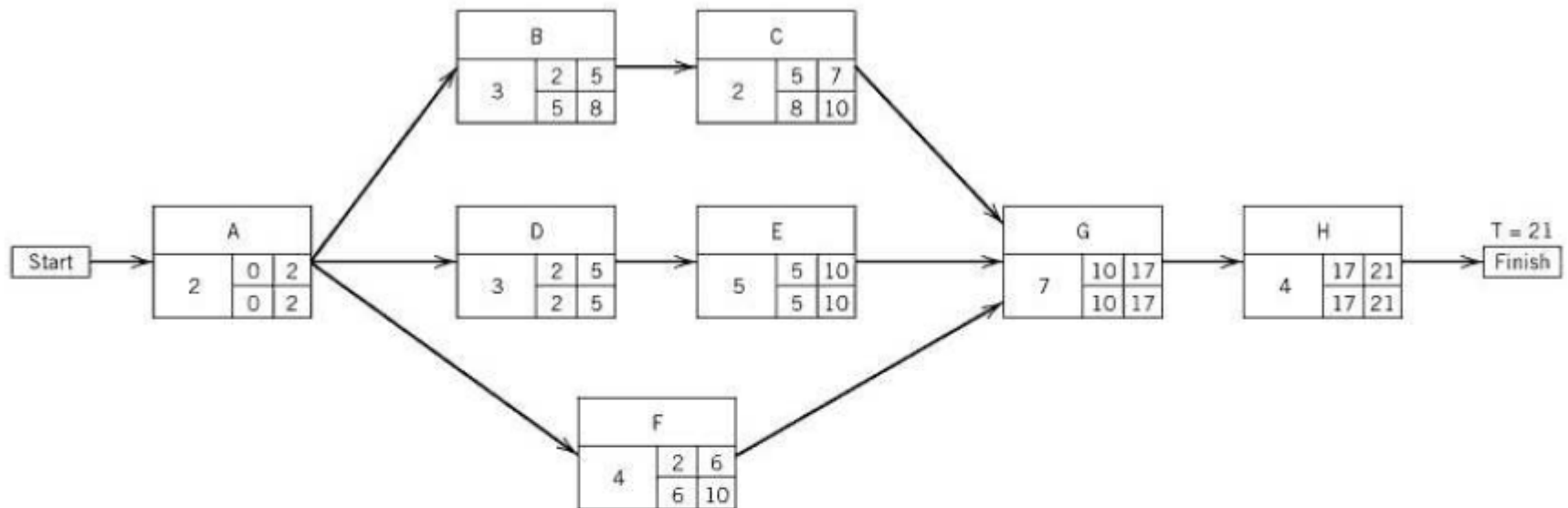
P: Team Participant

2.5.5 Contingency Diagram

The contingency diagram formally known as process decision program chart, maps conceivable events and contingencies that might occur during implementation. It particularly is useful when the project being planned consists of unfamiliar tasks.

2.5.6 Activity Network Diagram

The activity network diagram is used to develop a work schedule for the facilities design effort. This diagram is synonymous to the critical path method (CPM) graph. It can also be replaced by a Gantt chart.



Legend

- A: Schedule shutdown periods for equipment movement and installation
- B: Interview, evaluate, select, and hire new employees
- C: Train new employees
- D: Interview, evaluate, and select equipment vendors
- E: Order equipment
- F: Interview, evaluate, select, and hire construction contractors
- G: Meet to review installation plan (facilities design team, contractors, new employees, vendor representatives, and management)
- H: Executive installation plan and test system

Activity			
Di	ES	EF	
	LS	LF	

Di: Activity duration
 ES: Early start
 EF: Early finish
 LS: Late start
 LF: Late finish

2.5.7 Prioritization Matrix

In developing facilities design alternatives it is important to check

(a) Layout characteristics

- Total distance travelled
- Manufacturing floor visibility
- Overall aesthetics of the layout
- Ease of adding future business

(b) Material handling equipment

- Use of current material handling equipment
- Investment requirements on new equipment
- Space and people requirements

(c) Unit load implied

- Impact on WIP levels
- Space requirements
- Impact on material handling equipment

(d) Storage strategies

- Space and people requirements
- Impact on material handling equipment
- Human factor risk

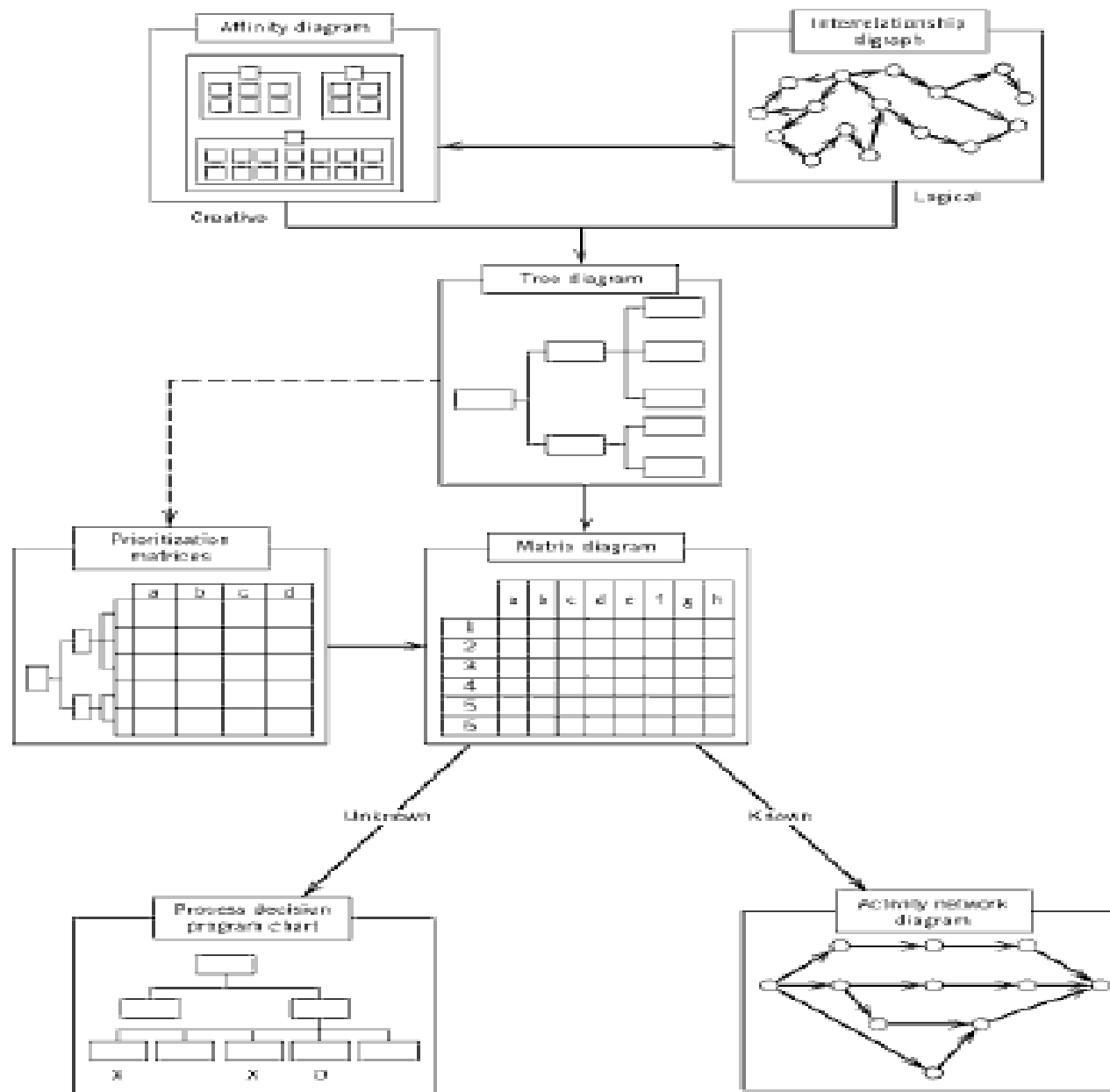
(e) Overall building impact

- Estimated cost of the alternative
- Opportunities for new business.

The prioritization matrix can be used to judge the relative importance of each criterion as compared to each other.

	Criteria											Row totals (%)
	A	B	C	D	E	F	G	H	I	J	K	
A	1	5	10	5	1	1	1	1	1	5	1	32 (9.9)
B	1/5	1	5	1/5	1/5	1/10	1/5	1/5	1/10	1/5	1/5	7.6 (2.4)
C	1/10	1/5	1	1/10	1/10	1/10	1/5	1/5	1/10	1/10	1/10	2.3 (0.7)
D	1/5	5	10	1	1/5	1/5	1/5	1/5	1/10	1/5	1/10	17.4 (5.4)
E	1	5	10	5	1	1	5	5	1/5	1	1/5	34.4 (10.7)
F	1	10	10	5	1	1	5	5	1	1	1	41 (12.7)
G	1	5	5	5	1/5	1/5	1	5	1/5	1/5	1/58	23 (7.1)
H	1	5	5	5	1/5	1/5	5	1	1/10	1/5	1/5	22.9 (7.1)
I	1	10	10	10	5	1	5	10	1	1	5	59 (18.3)
J	1/5	5	10	5	1	1	5	5	1	1	5	39.2 (12.2)
K	1	5	10	10	5	1	5	5	1/5	1/5	1	43.4 (13.5)
Column total	7.7	56.2	86	51.3	14.9	6.8	32.6	37.6	5	10.1	14	322.2
												Grand total

Prioritization matrix for the evaluation of facilities design alternatives



Logical application sequence of the seven management and planning tool

CHAPTER 3

FLOW, SPACE, AND ACTIVITY RELATIONSHIPS

- In determining the requirements of a facility, three important considerations are flow, space, and activity relationships.
 - **Flow** depends on lot sizes, unit load sizes, material handling equipment and strategies, layout arrangement, and building configuration.
 - **Space** is a function of lot sizes, storage system, production equipment type and size, layout arrangement, building configuration, housekeeping and organization policies, material handling equipment, and office, cafeteria, and restroom design.
 - **Activity relationships** are defined by material or personnel flow, environmental considerations, organizational structure, continuous improvement methodology (teamwork activities), control issues, and process requirements.